

Scalability & Future Plan

This document outlines the growth trajectory of OptiCrop beyond v1.0, covering planned phases, technical scalability, and long-term impact goals.

Phase Roadmap

Phase	Timeline	Key Features	Priority
v1.0 (Current)	Month 1-4	ML prediction, Flask UI, 22 crops, confidence score	Released
v1.1	Month 5-6	Hindi/Marathi language toggle (Flask-Babel)	High
v1.2	Month 5-7	PDF export of recommendation; crop growing calendar	High
v2.0	Month 7-10	Soil Health Card image OCR auto-fill inputs	High
v2.1	Month 8-11	WhatsApp Bot integration via Twilio API	Medium
v2.2	Month 9-12	Progressive Web App (PWA) with offline cache	Medium
v3.0	Year 2	IoT soil sensor ingestion via MQTT	Long-term
v3.1	Year 2	Market price API show crop profitability forecast	Long-term
v3.2	Year 2+	Mobile app (Flutter) with GPS-based weather auto-fill	Long-term

Technical Scalability Plan

Database (Phase 2)

Replace CSV prediction log with PostgreSQL. Enables multi-tenant data, regional analytics, and machine learning on prediction patterns.

API Layer (Phase 2)

Expose a REST API (FastAPI) so the prediction engine can serve mobile apps, WhatsApp bots, and third-party agricultural platforms independently.

Model Retraining Pipeline (Phase 2)

Implement an MLflow-tracked retraining pipeline that ingests new labelled prediction logs quarterly, retrains the Random Forest, and hot-swaps model.pkl without server downtime.

Cloud Deployment (Phase 3)

Deploy on AWS EC2 (t3.medium) behind an Application Load Balancer with auto-scaling group. Target: 10,000 predictions/day across 5 states.

Impact Targets

Milestone	Target	Timeline
Beta users (farmers)	500 farmers in 2 districts	Phase 2 (Month 8)
Prediction accuracy	ε 99.5% with expanded data	Phase 2 model retrain
Crop classes	Expand to 40+ crops	Phase 3
Languages supported	5 (EN, HI, MR, TE, TA)	Phase 2 3
Countries	India + Bangladesh + Kenya	Phase 3 (Year 2)
Farmers reached	100,000+	Year 3